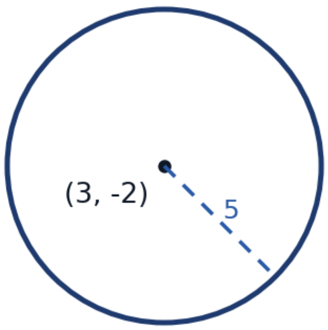
Circles and Parabolas

Circles

1 Write the standard-form equation of the circle with center $(3, -2)$ and radius 5.

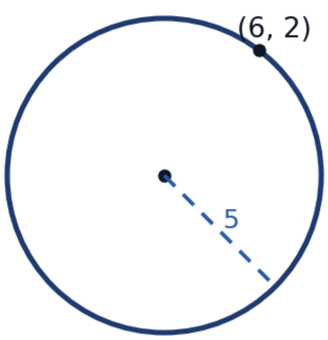
2 Consider the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.



a Write the equation in standard form by completing the square.

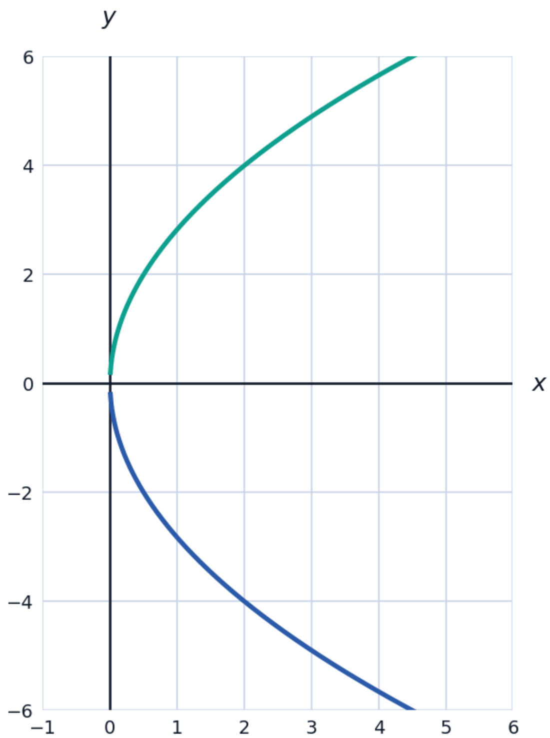
b State the center and radius.

3 Determine whether the point $(6, 2)$ lies inside, on, or outside the circle $(x-3)^2 + (y+2)^2 = 25$.



Parabolas with vertex at the origin

4 For the parabola $y^2 = 8x$, find:

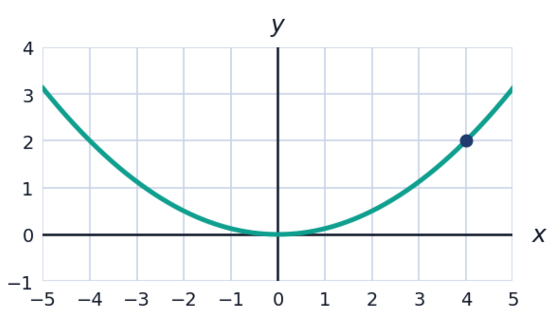


a the focus

b the directrix

c the direction of opening

5 A parabola has vertex at the origin, opens upward, and passes through the point $(4, 2)$.



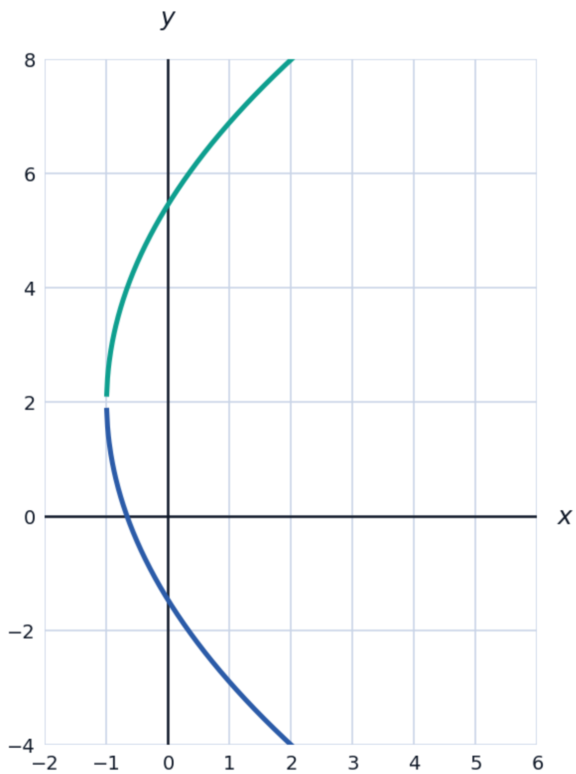
a Find its equation in the form $x^2 = 4py$.

b State the coordinates of the focus.

6 A satellite dish has a parabolic cross-section 2 m wide and 0.5 m deep. Taking the vertex at the origin opening upward, find the distance from the vertex to the focus (where the receiver is placed).

Parabolas with vertex not at the origin

7 For the parabola $(y-2)^2 = 12(x+1)$, find:



a the vertex

b the value of p

c the focus

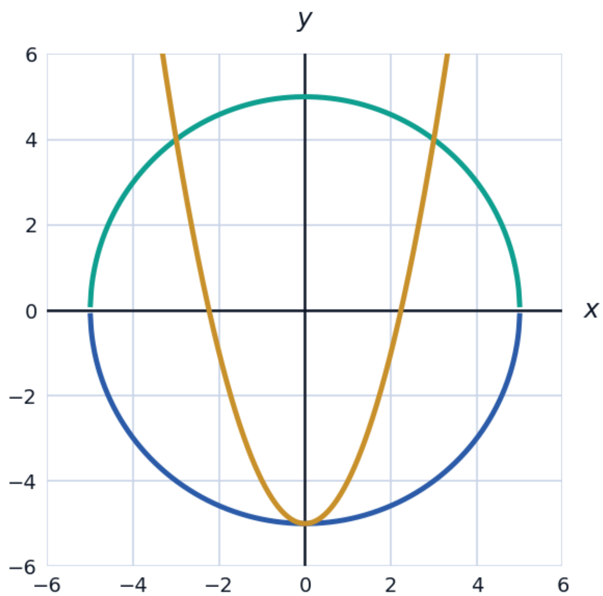
d the directrix

8 Write the equation of the parabola with vertex $(2, -3)$, opening downward, with $|p| = 1$.

9 Complete the square to write $y = x^2 - 6x + 5$ in vertex form, and state the vertex.

Mixed practice: circles and parabolas together

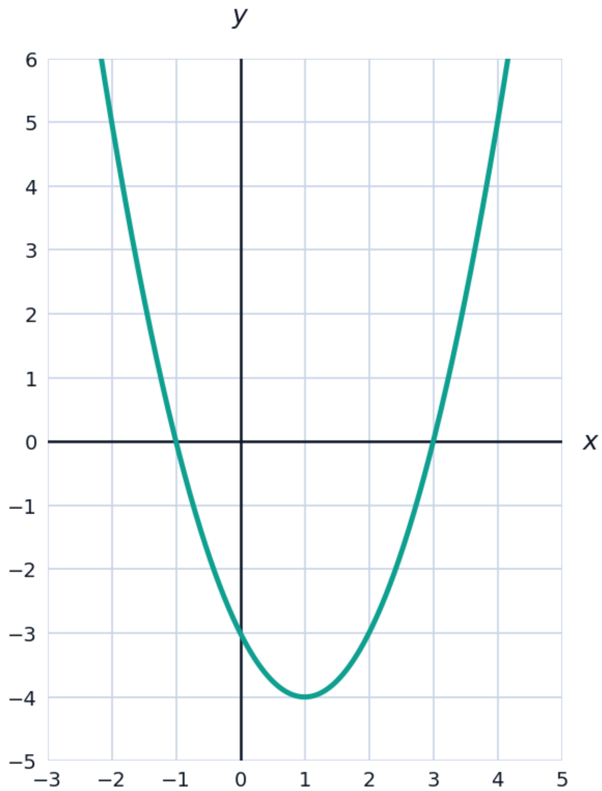
10 A circle and a parabola are given by $x^2 + y^2 = 25$ and $y = x^2 - 5$. Find their points of intersection.



11 A circular satellite orbit has the equation $x^2 + y^2 = (6400 + 500)^2$ (Earth radius plus altitude, in km). Find the radius of this orbit.

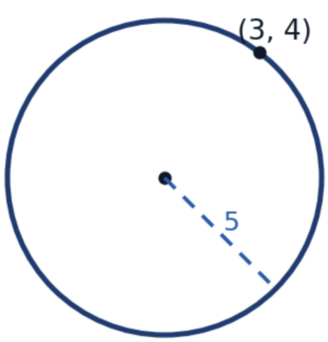
Graphing circles and parabolas

12 The parabola $y = x^2 - 2x - 3$ is shown. Use the graph to estimate its vertex and x-intercepts, then verify algebraically.



Tangent lines to circles

13 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.



The reflective property of parabolas

14 Explain the reflective property of a parabola that makes it useful for satellite dishes and headlights, in terms of the focus.

Finding a circle through three points

15 Explain the general strategy for finding the equation of a circle passing through three given (non-collinear) points, without fully solving a specific example.

Distance from a point to the focus (parabola definition)

16 State the focus-directrix definition of a parabola, and use it to verify that the point $(4, 4)$ lies on $y^2 = 4x$ (focus $(1, 0)$, directrix $x = -1$).

Systems involving circles

17 Find the points of intersection of the circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$.